care of this, we'll have to recast things in terms of position and canonical *momentum*, which does have a standard operator form. Rewriting the Hamiltonian in terms of canonical momentum is easy because

$$p = \frac{\partial L}{\partial \dot{x}} = \dot{x},$$

which allows us to write

$$H = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 x^2. \qquad (10.11)$$

That's the classical Hamiltonian. We can now turn it into a quantum mechanical equation by reinterpreting x and p as operators, defined by their action on $\psi(x)$. As we've done before, we'll use the boldface symbols, **X** and **P**, to distinguish our quantum operators from their classical counterparts, x and p. From previous lectures, we know exactly how these operators work. **X** just multiplies the wave function by the position variable:

$$\mathbf{X}|\psi(x)\rangle \qquad \Longrightarrow \qquad x\psi(x).$$

And **P** takes the same form it does for other one-dimensional problems:

$$\mathbf{P}|\psi(x)\rangle \qquad \Longrightarrow \qquad -i\hbar\frac{d}{dx}\psi(x).$$

Now, we can figure out the action of the Hamiltonian on a wave function by letting **P** act twice on the wave function. This is the same procedure we followed in Lecture 9. In other words,

$$\mathbf{H}|\psi(x)\rangle \qquad \Longrightarrow \qquad \frac{1}{2}\left(-i\hbar\frac{\partial}{\partial \mathrm{x}}\left(-i\hbar\frac{\partial\psi(\mathrm{x})}{\partial \mathrm{x}}\right)\right) + \frac{1}{2}\omega^2\mathrm{x}^2\psi(\mathrm{x}),$$